Independent audit of `onebody_scheme`: vacuum elasticity matching, sound-subtracted invariance, and the factor of four

Quantum Boussinesq solitons project

4 November 2026

Abstract

(c) We conduct an independent mathematical and numerical audit of `phase2/onebody_scheme.tex` and its three verification scripts (`onebody_scheme_continuum.py`, `onebody_scheme_dictionary.py`, and `onebody_scheme_edge.py`). All three scripts execute cleanly in the foreground and print `ALL CHECKS PASSED` without circularity. Independent mode sums across four continuum regularisations (sharp wavenumber cutoff, dimensional/analytic regularisation, heat-kernel damping, and the 2D Euclidean Feynman bubble integral) and two lattice methods confirm the vacuum sound-speed coefficients $s_{1,g} = -9/(4\pi)$ (good continuum), $s_{1,b,\mathrm{ac}} = +9/(4\pi)$ (formally continued bad continuum), and $s_{1,T} = +3/(4\pi)$ (quantum Toda lattice), refuting the assumption $s_{1,\mathrm{cont}} = 0$ in `audit_onebody.tex`. Matching the low-energy vacuum elastic modulus fixes the local effective Hamiltonian counterterm $\Delta\mathcal{H} = -\frac{3\hbar_B}{2\pi}u^2$ from homogeneous ($k = 0$) vacuum mode sums alone, with zero input from soliton data. Because the classical soliton canonical momentum satisfies $P_0 = v\int u_0^2\,\mathrm{d}x$, any counterterm $\Delta\mathcal{H} = \hbar_B\lambda u^2$ shifts both s_1 and the small-amplitude soliton slope $C = \lim_{k\to 0} F(k)/|P_0(k)|$ by exactly λ, shifting $C_{\mathrm{ac}} = 2/\pi$ by $\Delta C = -3/(2\pi)$ to $C_{\mathrm{matched}} = 1/(2\pi) = C_T$ and leaving the physical sound-subtracted soliton slope $C - s_1 = -1/(4\pi)$ strictly invariant. We also confirm the diagnosis of the failed line 383 assertion in `audit_onebody.py` as a coarse-grid $O((hy)^3)$ quadratic extrapolation truncation error (2.2547×10^{-4} at $y = 64$) on a true $h \to 0$ limiting identity. **(d)** The scheme explanation **(A)** stands for the leading $O(\hbar_B^0)$ coefficient of the stipulated formally continued trace; it does not construct a physical bad-equation fluctuation determinant, repair the real stable-band (Scheme A) resonance, or establish full finite-amplitude equivalence.

1 Scope, conventions, and script execution

Claim labels follow `AGENTS.md`: (a) proved in the literature, (b) conjectured or heuristic in the literature, (c) derived or verified in this project, and (d) a guess or unestablished physical identification. No external article, preprint, arXiv identifier, or DOI is cited here. The internal documents read in full for this audit are `AGENTS.md`, `README.md`, `phase2/report.tex`, `phase2/onebody_scheme.tex`, `phase2/audit_onebody.tex`, `phase2/onebody_dispersion_test.tex`, `phase2/onebody_toda_precision.tex`, `phase2/det_projected_oneloop.tex`, `phase2/frame_dictionary.tex`, `phase2/oneloop.tex`, and `phase2/audit_phase2i.tex`, together with their scripts and saved JSON files. All independent symbolic and 70-digit `mpmath` checks are implemented in `phase2/scripts/audit_onebody_scheme.py`, which imports no project script, verifies SHA-256 hashes of all audited files, and ends by printing `ALL CHECKS PASSED`.

(c) (`audit_onebody_scheme.py`) In the project's fixed conventions, $u = \phi_x$, $\pi = \phi_t$, $\Omega = \int \mathrm{d}\pi \wedge \mathrm{d}\phi$, and the good ($g$) and bad ($b$) Hamiltonian densities are

$$\mathcal{H}_g = \tfrac{1}{2}\pi^2 + \tfrac{1}{2}u^2 - u^3 + \tfrac{1}{2}u_x^2, \qquad \mathcal{H}_b = \tfrac{1}{2}\pi^2 + \tfrac{1}{2}u^2 + u^3 - \tfrac{1}{2}u_x^2, \tag{1}$$

generating $u_{tt} = u_{xx} \mp u_{xxxx} \mp 3(u^2)_{xx}$. The one-soliton profile $u_0(x,t) = \frac{k^2}{2}\operatorname{sech}^2\frac{k(x-vt)}{2}$ has $k = \sin\alpha$, $v_g = \epsilon\sqrt{1-k^2}$ (good) and $k = \sinh\beta$, $v_b = \epsilon\sqrt{1+k^2}$ (bad), with integral moments

$$\int_{-\infty}^{\infty} u_0\,\mathrm{d}x = 2k, \qquad \int_{-\infty}^{\infty} u_0^2\,\mathrm{d}x = \frac{2}{3}k^3, \qquad \int_{-\infty}^{\infty} u_0^3\,\mathrm{d}x = \frac{4}{15}k^5, \qquad \int_{-\infty}^{\infty} u_{0,x}^2\,\mathrm{d}x = \frac{2}{15}k^5. \tag{2}$$

Throughout, $\hbar_B W(U)$ denotes the $O(\hbar_B)$ homogeneous vacuum energy density on a constant background strain $u = U$, $c_{\text{sound}} = 1 + \hbar_B s_1 + O(\hbar_B^2)$ denotes the quantum sound speed in fixed classical Boussinesq units, and $\hbar_B F(k) = E_1\big|_P$ denotes the $O(\hbar_B)$ one-soliton energy correction at fixed canonical momentum P, with small-amplitude slope $C := \lim_{k\to 0^+} F(k)/|P_0(k)|$.

(c) Foreground execution of the audited scripts from the repository root gave:

1. `python3 phase2/scripts/onebody_scheme_continuum.py`: exited 0 in 6 s, printed `ALL CHECKS PASSED`.
2. `python3 phase2/scripts/onebody_scheme_dictionary.py`: exited 0 in 57 s, printed `ALL CHECKS PASSED`.
3. `python3 phase2/scripts/onebody_scheme_edge.py`: exited 0 in 312 s, printed `ALL CHECKS PASSED`.
4. `python3 phase2/scripts/audit_onebody.py`: exited 1 at line 383 with `AssertionError` on `max_cv_id_diff < 3e-6`, confirming the failure reported in `onebody_scheme.tex`.

2 Point 1: Independent recomputation of the vacuum sound-speed shifts

2.1 Good continuum vacuum: $s_{1,g} = -9/(4\pi)$ across four regularisations

(c) (`audit_onebody_scheme.py`) Expanding $\mathcal{H}_g$ around a homogeneous background strain $\phi_0(x) = Ux$ ($u = U + \eta_x$, $\pi = \eta_t$, with $U < 1/6$) gives

$$\mathcal{H}_g(U+\eta_x) = \left(\tfrac{1}{2}U^2 - U^3\right) + (U - 3U^2)\eta_x + \tfrac{1}{2}\eta_t^2 + \tfrac{1}{2}(1-6U)\eta_x^2 + \tfrac{1}{2}\eta_{xx}^2 - \eta_x^3, \tag{3}$$

so small fluctuations have dispersion $\omega_g(q;U) = |q|\sqrt{q^2 + m^2(U)}$ with $m^2(U) = 1 - 6U$, and classical sound speed squared $c_{s,g,0}^2(U) = 1 - 6U$. Because the cubic interaction $-\phi_x^3$ depends only on ϕ_x and not on $\phi_t = \pi$, every external leg of the one-loop 1PI self-energy $\Sigma_{\phi\phi}(p_t, p_x)$ carries a spatial momentum factor ip_x. Thus $\Sigma_{\phi\phi}(p_t, p_x) = p_x^2\widetilde{\Sigma}(p_t, p_x)$ has $\partial_{p_t}^2\Sigma_{\phi\phi}(0,0) = 0$: the canonical kinetic term $\frac{1}{2}\pi^2$ receives no one-loop wave-function renormalisation, and the thermodynamic sound speed squared at $U = 0$ is $c_{s,g}^2 = 1 + \hbar_B W_g''(0) + O(\hbar_B^2)$, giving $s_{1,g} = \frac{1}{2}W_g''(0)$ **(c)** .

We compute $W_g(U)$ and $W_g''(0)$ by four independent regularisations (`audit_onebody_scheme.py`):

1. **Sharp wavenumber cutoff $|q| \le \Lambda$:** $W_{g,\Lambda}^{\text{bare}}(U) = \frac{1}{2\pi}\int_0^\Lambda q\sqrt{q^2 + 1 - 6U}\,\mathrm{d}q = \frac{(\Lambda^2+1-6U)^{3/2} - (1-6U)^{3/2}}{6\pi}$. In the specified Wick normal ordering $:u^3: := u^3 - 3\langle u^2\rangle_{0,\Lambda}u$ with $\langle u^2\rangle_{0,\Lambda} = \frac{\hbar_B}{2\pi}(\sqrt{\Lambda^2+1} - 1)$, passing to $:\mathcal{H}_g:$ adds the counterterm density $+3\langle u^2\rangle_{0,\Lambda}U = -U\partial_U W_{g,\Lambda}^{\text{bare}}(0)\hbar_B$. Subtracting the $U = 0$ vacuum energy and this exact linear tadpole gives a UV-convergent $\Lambda \to \infty$ limit:

$$W_g(U) = \lim_{\Lambda\to\infty}\left[W_{g,\Lambda}^{\text{bare}}(U) - W_{g,\Lambda}^{\text{bare}}(0) - U\partial_U W_{g,\Lambda}^{\text{bare}}(0)\right] = -\frac{(1-6U)^{3/2} - 1 + 9U}{6\pi}. \tag{4}$$

2. **Dimensional / analytic (Riesz) regularisation:** Using $I_\varepsilon(M^2) = \frac{1}{2\pi}\int_0^\infty q(q^2 + M^2)^{\frac{1}{2}-\varepsilon}\,\mathrm{d}q = -\frac{(M^2)^{\frac{3}{2}-\varepsilon}}{4\pi(\frac{3}{2}-\varepsilon)}$ (continued from $\operatorname{Re}\varepsilon > 3/2$) and subtracting $I_\varepsilon(1)$ and the corresponding normal-ordering tadpole $-6UI_\varepsilon'(1)$ cancels the regulator pole and yields (4) at $\varepsilon \to 0$ **(c)** .

3. **Heat-kernel damping and convergent second derivative:** Differentiating twice with respect to U inside the mode integral gives a manifestly UV-convergent integral in $1+1$ dimensions:

$$W_g''(U) = -\frac{9}{2\pi}\int_0^\infty \frac{q\,\mathrm{d}q}{(q^2+1-6U)^{3/2}} = -\frac{9}{2\pi\sqrt{1-6U}}, \qquad s_{1,g} = \frac{1}{2}W_g''(0) = -\frac{9}{4\pi}. \tag{5}$$

With exponential damping $e^{-\delta q^2}$, the integral is $-\frac{9}{2\pi}[1-\sqrt{\pi\delta}e^{\delta}\operatorname{erfc}(\sqrt{\delta})] \to -\frac{9}{2\pi}$ as $\delta \to 0^+$. Furthermore, rationalising the subtracted integrand on $q \in [0,\infty)$ as

$$q\left(\sqrt{q^2+1-6U} - \sqrt{q^2+1} + \frac{3U}{\sqrt{q^2+1}}\right) = -\frac{18U^2 q}{\sqrt{q^2+1}\left(\sqrt{q^2+1-6U}+\sqrt{q^2+1}\right)^2} \tag{6}$$

eliminates catastrophic cancellation at large q and reproduces (4) by direct infinite-line quadrature at $U \in \{0.02, -0.03, 0.05\}$ to $< 10^{-65}$ (`audit_onebody_scheme.py`).

4. **2D Euclidean Feynman bubble diagram:** Evaluating the static one-loop bubble diagram with two cubic $-u^3$ vertices gives

$$W_g''(0) = -18\int_{-\infty}^{\infty}\frac{\mathrm{d}q}{2\pi}\int_{-\infty}^{\infty}\frac{\mathrm{d}\omega_E}{2\pi}\frac{q^4}{\left(\omega_E^2+q^2(q^2+1)\right)^2} = -\frac{9}{2\pi}\int_0^\infty \frac{q\,\mathrm{d}q}{(q^2+1)^{3/2}} = -\frac{9}{2\pi}. \tag{7}$$

(c) This confirms `onebody_scheme.tex` Eqs. (18)–(20) and refutes `audit_onebody.tex`'s claim that $s_{1,\mathrm{cont}} = 0$ in the bare-normalised (normal-ordered only) continuum theory.

2.2 Continued bad vacuum ($s_{1,b,\mathrm{ac}} = +9/(4\pi)$) and Toda vacuum ($s_{1,T} = +3/(4\pi)$)

(c) /**(d)** (`audit_onebody_scheme.py`) On the bad branch $\mathcal{H}_b$, the formal continuation $F_{\mathrm{ac}}(k) = -iF_g(ik)$ used for the soliton trace corresponds in the scaling family $\mathcal{L}_\sigma$ ($\sigma = c^2$ at $c = i$) to $x_g = -ix_b$ and $U_g = -U_b$, so $-i\int \mathrm{d}x_g\, W_g(U_g) = -\int \mathrm{d}x_b\, W_g(-U_b)$, giving

$$W_{b,\mathrm{ac}}(U) = -W_g(-U) = \frac{(1+6U)^{3/2}-1-9U}{6\pi}, \qquad W_{b,\mathrm{ac}}''(0) = +\frac{9}{2\pi}, \qquad s_{1,b,\mathrm{ac}} = +\frac{9}{4\pi}. \tag{8}$$

(c) (`audit_onebody_scheme.py`) We also verify (8) directly from the bad dispersion $\omega_b(q;U) = |q|\sqrt{1+6U-q^2}$ on its strain-dependent real stable band $0 \le |q| \le \sqrt{1+6U}$: integrating $\frac{1}{2\pi}\int_0^{\sqrt{1+6U}} q\sqrt{1+6U-q^2}\,\mathrm{d}q = \frac{(1+6U)^{3/2}}{6\pi}$ and subtracting the $U=0$ constant $\frac{1}{6\pi}$ and the bad normal-ordering tadpole $-3\langle u^2\rangle_0^{\mathrm{band}}U/\hbar_B = -\frac{3U}{2\pi}$ yields exact agreement with (8). (Differentiating under the integral sign on a fixed interval $[0,1]$ instead requires the Hadamard finite part $\mathrm{FP}\int_0^1 q(1-q^2)^{-3/2}\,\mathrm{d}q = -1$ because the band edge $\sqrt{1+6U}$ moves with U.) As `onebody_scheme.tex` rightly states, the algebraic continuation (8) is **(c)** , whereas interpreting it as a physical bad-equation vacuum despite the unstable modes $|q| > 1$ is **(d)** .

(c) (`audit_onebody_scheme.py`) For the Toda lattice at reference spacing a, $b = e^{-a/2}$, and $\hbar_T = h = 3b\hbar_B$, a uniform Boussinesq strain $U = \partial_{x_B}\phi_B$ deforms the lattice spacing to $r = a - 6U$, multiplying the harmonic lattice dispersion $\omega_T(K;r) = 2be^{3U}|\sin(K/2)|$ by e^{3U} across the fixed Brillouin zone $K \in [-\pi,\pi]$. The one-loop zero-point energy per site is $e_{1,T}(r) = \frac{h}{4\pi}\int_0^{2\pi} 2be^{3U}\sin(K/2)\,\mathrm{d}K = \frac{2hb}{\pi}e^{3U}$. Converting to Boussinesq energy density via $\Delta E_B = \Delta E_T/(6\sqrt{3}b^2)$ and $\mathrm{d}x_B = 2\sqrt{3}\,\mathrm{d}n$ gives $\hbar_B W_{T,\mathrm{bare}}^B(U) = \frac{e_{1,T}(a-6U)}{36b^2} = \hbar_B\frac{e^{3U}}{6\pi}$. Subtracting the $U=0$ constant and linear pressure term yields

$$W_T^B(U) = \frac{e^{3U}-1-3U}{6\pi}, \qquad (W_T^B)''(0) = \frac{3}{2\pi}, \qquad s_{1,T} = \frac{1}{2}(W_T^B)''(0) = +\frac{3}{4\pi}, \tag{9}$$

which agrees with the non-perturbative Bethe scaling identity $v_s(a,h) = \frac{a}{2}(q - h\partial_h q|_a) = ab + \frac{ah}{4\pi} + O(h^2\ln h)$ **(c)** .

Why $s_{1,b,\mathrm{ac}} = 9/(4\pi)$ differs by a factor of 3 from $s_{1,T} = 3/(4\pi)$ (c) (`audit_onebody_scheme.py`). In the Toda lattice, the classical potential in Boussinesq units is $V^B_{\mathrm{Toda}}(U) = \frac{e^{6U}-1-6U}{36} = \frac{1}{2}U^2 + U^3 + \frac{3}{2}U^4 + O(U^5)$, and a uniform strain U rescales $\omega_T(K;U) = e^{3U}\omega_T(K;0)$ on a *strain-independent* Brillouin zone $K \in [-\pi,\pi]$, producing $W^B_{T,\mathrm{bare}}(U) \propto e^{3U} = 1 + 3U + \frac{9}{2}U^2 + \cdots$. In the truncated continuum bad Hamiltonian $\mathcal{H}_b$, the quartic vertex $\frac{3}{2}U^4$ is omitted and $-q^4$ inside $\omega_b(q;U) = |q|\sqrt{1+6U-q^2}$ is not multiplied by e^{6U}; rescaling $q = \sqrt{1+6U}\,\tilde{q}$ pulls $(1+6U)^{1/2}$ from the frequency and $(1+6U)$ from the measure $q\,\mathrm{d}q$, producing $W^{\mathrm{bare}}_{b,\mathrm{ac}}(U) \propto (1+6U)^{3/2} = 1 + 9U + \frac{27}{2}U^2 + \cdots$, whose quadratic curvature is three times larger ($\frac{27}{2}$ versus $\frac{9}{2}$).

3 Points 2 and 3: Non-circularity of $\Delta\mathcal{H}$ and its exact effect on the soliton

Point 2: Non-circularity of $\Delta\mathcal{H} = -\frac{3\hbar_B}{2\pi}u^2$ (c) (`audit_onebody_scheme.py`). Subtracting (8) from (9) gives the homogeneous vacuum energy-density mismatch between the Toda lattice and the continued continuum prescription:

$$W^B_T(U) - W_{b,\mathrm{ac}}(U) = \frac{e^{3U} - (1+6U)^{3/2} + 6U}{6\pi} = -\frac{3}{2\pi}U^2 + \frac{3}{\pi}U^3 - \frac{9}{2\pi}U^4 + O(U^5). \tag{10}$$

Matching the vacuum elastic modulus (the $O(U^2)$ coefficient) in the same reference coordinates uniquely fixes the local quadratic counterterm in the continuum effective Hamiltonian:

$$\Delta\mathcal{H} = \hbar_B \lambda u^2, \qquad \lambda = s_{1,T} - s_{1,b,\mathrm{ac}} = \frac{3}{4\pi} - \frac{9}{4\pi} = -\frac{3}{2\pi}. \tag{11}$$

Audit of `onebody_scheme.tex` and `onebody_scheme_continuum.py` (lines 18–57) confirms that $\lambda = -\frac{3}{2\pi}$ is derived *exclusively* from the homogeneous ($k = 0$) vacuum mode sums $W^B_T(U)$ and $W_{b,\mathrm{ac}}(U)$: neither the soliton profile $u_0(x)$, nor the continuum soliton coefficient $C_{\mathrm{ac}} = 2/\pi$, nor the Toda Fermi-edge coefficient $C_T = 1/(2\pi)$ is used anywhere to choose λ. The linear $O(U)$ difference between $W^B_{T,\mathrm{bare}}$ and $W^{\mathrm{bare}}_{b,\mathrm{ac}}$ is a background lattice-pressure term $\propto M = \int u\,\mathrm{d}x = 2k$, which is already fixed to zero in the soliton dispersion by subtracting the Fermi-edge chemical potential $E(q)$ on the Toda side and by the equal-mode-count endpoint plus Wick tadpole on the continuum side **(c)** .

Point 3: Why adding $\Delta\mathcal{H} = \hbar_B\lambda u^2$ shifts C_{ac} by exactly $\lambda = -\frac{3}{2\pi}$ (c) (`audit_onebody_scheme.py`). First, because $\Delta\mathcal{H} = \hbar_B\lambda\phi_x^2$ contains no time derivative ϕ_t, the canonical conjugate field $\pi = \partial\mathcal{L}/\partial\phi_t = \phi_t$ and the canonical field momentum functional $P[\phi,\pi] = -\int \pi\phi_x\,\mathrm{d}x$ are unchanged. Second, since the classical soliton (ϕ_0,π_0) is a stationary point of $H_0 - vP$ ($\delta H_0 = v\,\delta P$), any $O(\hbar_B)$ deformation of the soliton profile at fixed canonical momentum ($\delta P = 0$) contributes zero to the $O(\hbar_B)$ energy by first-order perturbation theory. Thus the shift in $F(k) = E_1|_P/\hbar_B$ is simply $\Delta F(k) = \lambda\int_{-\infty}^{\infty} u_0(x)^2\,\mathrm{d}x$.

Third, for *any* travelling-wave soliton $\phi_0(x,t) = \Phi_0(x-vt)$, we have $\pi_0 = -vu_0$, so the classical canonical momentum is identically

$$P_0 = -\int_{-\infty}^{\infty}\pi_0\phi_{0,x}\,\mathrm{d}x = v\int_{-\infty}^{\infty}u_0(x)^2\,\mathrm{d}x \quad\Longrightarrow\quad \Delta F(k) = \lambda\int_{-\infty}^{\infty}u_0(x)^2\,\mathrm{d}x = \lambda\,\frac{P_0(k)}{v(k)} = \frac{2\lambda}{3}k^3. \tag{12}$$

Since $|v(k)| = \sqrt{1\pm k^2} \to 1$ as $k \to 0^+$, dividing (12) by $|P_0(k)|$ gives

$$\Delta C = \lim_{k\to 0^+}\frac{\Delta F(k)}{|P_0(k)|} = \lim_{k\to 0^+}\frac{\lambda}{|v(k)|} = \lambda = -\frac{3}{2\pi}, \qquad C_{\mathrm{matched}} = C_{\mathrm{ac}} + \lambda = \frac{2}{\pi} - \frac{3}{2\pi} = \frac{1}{2\pi} = C_T. \tag{13}$$

Higher-order local terms in (10) such as u^3, and dispersive matching terms such as u_x^2, have soliton integrals $\int u_0^3\,\mathrm{d}x = \frac{4}{15}k^5$ and $\int u_{0,x}^2\,\mathrm{d}x = \frac{2}{15}k^5$ of order $O(k^5) = O(|P_0|^{5/3})$, so their contribution to $\lim_{k\to 0}F(k)/|P_0(k)|$ vanishes identically **(c)** .

4 Points 4 and 5: Naturalness of the renormalisation condition and the good equation

Point 4: Naturalness and invariance of the sound-subtracted slope $C - s_1$ (c) (`audit_onebody_scheme.py`). In a gapless 1+1-dimensional effective field theory, $u^2 = \phi_x^2$ is the marginal elastic operator already present in the tree-level Hamiltonian $\frac{1}{2}\pi^2 + \frac{1}{2}u^2$. Because adding $\Delta\mathcal{H} = \hbar_B \lambda u^2$ shifts the vacuum sound-speed coefficient $s_1 \mapsto s_1 + \lambda$ and the small-amplitude soliton slope $C \mapsto C + \lambda$ by the *exact same constant* λ, the physical soliton slope relative to the quantum sound cone,

$$C_{\rm rel} := C - s_1 = \lim_{P\to 0} \frac{1}{\hbar_B}\left(\frac{E(P)}{|P|} - c_{\rm sound}\right), \tag{14}$$

is an *exact renormalisation-scheme invariant* (independent of λ) **(c)** . Without adding any counterterm, the continued continuum trace and the Toda lattice already have the *identical* sound-subtracted slope:

$$C_{\rm ac} - s_{1,b,\rm ac} = \frac{2}{\pi} - \frac{9}{4\pi} = -\frac{1}{4\pi} = \frac{1}{2\pi} - \frac{3}{4\pi} = C_T - s_{1,T}. \tag{15}$$

Depending on the coordinate/renormalisation convention chosen for the background sound speed, there are three natural choices of λ on the continued bad branch (`audit_onebody_scheme.py`):

Renormalisation scheme	λ	Vacuum s_1	Soliton C	Invariant $C - s_1$
1. Bare normal-ordered ($:u^3:$ only)	0	$+9/(4\pi)$	$+2/\pi$	$-1/(4\pi)$
2. On-shell continuum sound ($c_{s,\rm ren} = 1$)	$-9/(4\pi)$	0	$-1/(4\pi)$	$-1/(4\pi)$
3. Toda-matched EFT (inherited Toda units)	$-3/(2\pi)$	$+3/(4\pi)$	$+1/(2\pi)$	$-1/(4\pi)$

Standard QFT/EFT matching between a lattice UV completion (analyzed in fixed bare time coordinates where $c_{s,T} = 1 + \frac{3\hbar_B}{4\pi}$) and a continuum EFT requires matching their low-energy vacuum elastic moduli (Scheme 3 above), or equivalently comparing the scheme-invariant physical observable $C - s_1 = -1/(4\pi)$ (Scheme 2). The factor of 4 between $C_{\rm ac} = 2/\pi$ and $C_T = 1/(2\pi)$ (and the unmatched ratio $-8 = C_{\rm ac}/(C_T - s_{1,T})$ in `audit_onebody.tex`) arose solely from comparing Scheme 1 on the continuum side ($s_{1,b,\rm ac} = +9/(4\pi)$) against Scheme 3 on the Toda side ($s_{1,T} = +3/(4\pi)$) without matching the vacuum sound speeds **(c)** .

Point 5: The good equation (c) /(d) (`audit_onebody_scheme.py`). The good Boussinesq equation is linearly well-posed ($\omega_g(q;U) = |q|\sqrt{q^2 + 1 - 6U} \in \mathbb{R}$ for all $q \in \mathbb{R}$ and $U < 1/6$) and is not the real scaling limit of the Toda lattice, so Toda does not force the bad-branch lattice counterterm onto the good equation. On the good branch, the conditional soliton trace $F_g(k) = \frac{\sqrt{3}}{\pi}(\frac{3}{4} - k^2)\,\mathrm{artanh}\,\frac{2k}{\sqrt{3}} - \frac{3k}{2\pi}$ and the convergent vacuum bubble (5) give:

- **Bare normal-ordered scheme ($\lambda_g = 0$):** $s_{1,g} = -\frac{9}{4\pi}$ and $C_g^{\rm bare} = -\frac{2}{\pi}$, with sound-subtracted slope $C_g^{\rm bare} - s_{1,g} = -\frac{2}{\pi} - (-\frac{9}{4\pi}) = +\frac{1}{4\pi}$ **(c)** .
- **Natural on-shell acoustic renormalisation scheme ($c_{s,g,\rm ren} = 1$, i.e. $s_{1,g}^{\rm ren} = 0$):** Fixing the renormalised vacuum sound speed to 1 requires the finite elastic counterterm $\Delta\mathcal{H}_g^{\text{on-shell}} = +\frac{9\hbar_B}{4\pi}u^2$ ($\lambda_g = -s_{1,g} = +\frac{9}{4\pi}$), which shifts the soliton coefficient to

$$C_g^{\text{on-shell}} = C_g^{\rm bare} + \lambda_g = C_g^{\rm bare} - s_{1,g} = -\frac{2}{\pi} + \frac{9}{4\pi} = +\frac{1}{4\pi}. \tag{16}$$

Thus C_g is $-\frac{2}{\pi}$ in the bare normal-ordered scheme and $+\frac{1}{4\pi}$ in the on-shell renormalised-sound-speed scheme; in every scheme the physical slope relative to the quantum sound cone is $C_g - s_{1,g} = +\frac{1}{4\pi}$ **(c)** .

5 Point 6: Diagnosis of `audit_onebody.py`'s failing assertion and text errors

(c) (`audit_onebody_scheme.py`) We confirm `onebody_scheme.tex`'s diagnosis of line 383 of `audit_onebody.py`:

1. **Why line 383 fails on the saved data:** The relation $C_v(y) = s_{1,T} + \frac{1}{2}y\big(u'(y) - 1\big)$ is an exact $h \to 0$ *Fermi-edge limit* identity, verified to $< 10^{-45}$ by independent 50-digit Talbot and de Hoog Laplace inversions of $\mathcal{L}[F] = \sqrt{\pi}g(s)/s^2$ and $\mathcal{L}[f] = \sqrt{\pi}g(s)/s$ (`audit_onebody_scheme.py`). At finite h, however, the exact Bethe scaling identity leaves a discrepancy $R_h(y) = C_{v,h}(y) - s_{1,h} - \frac{y}{2}(\partial_y u_h - \frac{\sinh\kappa}{\kappa}) = O(hy^2)$ because $\kappa^2 = hy/b$ is finite ($\kappa \approx 1.45$ at $y = 64$, $h = 0.02$). In `onebody_toda_results.json` and `frame_analysis.json`, the central $h \to 0$ extrapolations use a 3-point Lagrange quadratic on $h \in \{0.02, 0.01, 0.005\}$ with weights $(\frac{1}{3}, -2, \frac{8}{3})$, which cancels $O(hy)$ and $O((hy)^2)$ but leaves an $O((hy)^3)$ truncation error that grows as y^3:

$$R_{\text{quad3}}(64) = \tfrac{1}{3}R_{0.02}(64) - 2R_{0.01}(64) + \tfrac{8}{3}R_{0.005}(64) = -2.25467349 \times 10^{-4}, \qquad (17)$$

   exceeding `audit_onebody.py`'s 3×10^{-6} bound at $y = 16, 32, 64$ (-5.10×10^{-6}, -3.48×10^{-5}, -2.25×10^{-4}), while remaining well inside the combined fit-sensitivity spread (1.101×10^{-3} at $y = 64$). On the finer `onebody_prec_scan.json` grid down to $h = 0.0025$, a 7-point quartic extrapolation reduces the $y = 64$ residual by 240× to $-9.380 \times 10^{-7} < 3 \times 10^{-6}$ (and -2.268×10^{-7} for an 8-point quintic, while a 6-point cubic gives -5.578×10^{-6}) **(c)** .

2. **Four additional algebraic errors in `audit_onebody.tex` confirmed (`audit_onebody_scheme.py`):** (i) Theorem 3 line 285 prints $b = \pi/a$ instead of $b = e^{-a/2}$; (ii) line 298 prints $\lim_{h\to 0}(V/D - v_s)/h = \frac{a}{6}y(u'(y) - 1)$ instead of $\frac{a}{6}yu'(y)$ (the $-\frac{a}{6}y$ term comes from subtracting the classical soliton velocity excess $ab(\frac{\sinh\kappa}{\kappa} - 1)/h$, not v_s); (iii) Eq. (24) prints $+\frac{3e_0}{2y^2}$ in $C_E(y)$ instead of $+\frac{3e_0}{y^2}$ because $\frac{3}{4}y^{-3/2}\int y^{3/2}(-2e_0y^{-3})\,\mathrm{d}y = +\frac{3e_0}{y^2}$; (iv) lines 197–199 misstate the bad stable-band resonance at $q_* = -\sqrt{3}/2$ (for $v = +1$) as $\omega_b(q_*) = -vq_*$ with $\omega_b'(q_*) = -1/2$, whereas in fact $\omega_b(q_*) = \frac{\sqrt{3}}{4} \neq \pm q_*$ and $\omega_b'(q_*) = \mathrm{sgn}(q_*)\frac{1-2q_*^2}{\sqrt{1-q_*^2}} = +1 = v$ (a stationary co-moving phase resonance $\omega_b'(q_*) = v$).

3. **Minor table spread discrepancy in `onebody_scheme.tex` caught (`audit_onebody_scheme.py`):** In the table at lines 94–98 of `onebody_scheme.tex`, the "combined fit spread" column prints 2.3259×10^{-5}, 8.4061×10^{-5}, and 2.9805×10^{-4} at $y = 8, 16, 32$ (which is $\Delta C_v + \frac{y}{2}\Delta u'$, omitting the sound-speed spread $\Delta s_{1,T} = 1.685 \times 10^{-8}$), whereas `onebody_scheme_edge.py` line 33 and its JSON include $\Delta s_{1,T}$, giving 2.3275×10^{-5}, 8.4078×10^{-5}, and 2.9806×10^{-4}. This 1.69×10^{-8} difference in the fourth significant digit of the diagnostic spread does not affect any conclusion.

6 Verdict table and failures recorded

Audited claim in `onebody_scheme`	Verdict	Independent audit finding (`audit_onebody_scheme.py`)
1. Vacuum sound-speed shifts: $s_{1,g} = -9/(4\pi)$, $s_{1,b,\text{ac}} = +9/(4\pi)$, $s_{1,T} = +3/(4\pi)$	Confirmed **(c)** (bad vacuum **(d)**)	Verified by 4 continuum regularisations (cutoff, dim-reg, heat-kernel, 2D Feynman bubble) and 2 Toda derivations. Refutes $s_{1,\text{cont}} = 0$ in `audit_onebody.tex`.

Audited claim in `onebody_scheme`	Verdict	Independent audit finding (`audit_onebody_scheme.py`)
2. Non-circularity of counterterm $\Delta\mathcal{H} = -\frac{3\hbar_B}{2\pi}u^2$	Confirmed **(c)**	Fixed purely by $O(U^2)$ matching of homogeneous ($k=0$) vacuum energy densities $W_T^B(U) - W_{b,\mathrm{ac}}(U)$; zero soliton input used.
3. Soliton shift $\Delta C = -3/(2\pi)$ and $C_{\mathrm{matched}} = 1/(2\pi) = C_T$	Confirmed **(c)**	Exact because $P_0 = v\int u_0^2\,\mathrm{d}x \implies \Delta F/\lvert P_0\rvert = \lambda/\lvert v\rvert \to \lambda$. Higher-order u^3, u_x^2 terms are $O(k^5)$ and vanish in $F/\lvert P_0\rvert$ as $k\to 0$.
4. Naturalness & sound-subtracted invariance $C - s_1 = -1/(4\pi)$	Confirmed **(c)**	$C - s_1 = -1/(4\pi)$ is invariant under $\Delta\mathcal{H} = \hbar_B\lambda u^2$ and agrees between Toda and continued continuum without tuning. Unmatched ratio -8 refuted as apples-to-oranges.
5. Good equation: no Toda-forced shift; C_g values	Confirmed **(c)** /**(d)**	Bare normal-ordered $C_g = -2/\pi$ ($s_{1,g} = -9/(4\pi)$); natural on-shell scheme ($c_{s,g,\mathrm{ren}} = 1$, $\lambda_g = +9/(4\pi)$) gives $C_g^{\text{on-shell}} = +1/(4\pi)$.
6. Diagnosis of `audit_onebody.py` line 383 failure	Confirmed **(c)**	Failure ($2.2547\times10^{-4} > 3\times10^{-6}$ at $y=64$) is an $O((hy)^3)$ 3-point quadratic truncation error; 7-point quartic gives -9.38×10^{-7}. Four text errors in `audit_onebody.tex` confirmed.
7. Real stable-band trace (Scheme A) F_{band}	Confirmed **(c)**	$F_{\mathrm{band}} = F_{\mathrm{ac}} - \frac{\sqrt{3}}{2}(k^2+\frac{3}{4})$ verified to $< 10^{-62}$; diverges as $-9\sqrt{3}/(16k^3)$ in $F/\lvert P_0\rvert$ and is not repaired by $\Delta\mathcal{H}$.
8. Scope of scheme explanation **(A)**	Confirmed **(c)** /**(d)**	**(A)** holds for the leading coefficient of the stipulated continued trace **(c)** ; full bad determinant, measure, and finite-k equivalence remain **(d)** .

Failures recorded.

1. **Unchanged `audit_onebody.py` line 383 failure (confirmed):** Running `python3 phase2/scripts/audit_onebody.py` fails at line 383 with maximum residual $2.2546734882\times 10^{-4}$ at $y=64$ (and fails at $y=16,32$). In `audit_onebody_scheme.py`, this failure and the four algebraic errors in `audit_onebody.tex` (plus its $s_{1,\mathrm{cont}}=0$ assumption and unmatched ratio -8) are encoded as explicit discrepancy assertions.

2. **Minor table spread omission in `onebody_scheme.tex` lines 95–98:** The text table in `onebody_scheme.tex` omitted $\Delta s_{1,T} = 1.685\times10^{-8}$ from the "combined fit spread" column at $y=8,16,32$, whereas `onebody_scheme_edge.py` and its JSON included it. Recorded and verified in `audit_onebody_scheme.py`.

3. **Development failures during construction of `audit_onebody_scheme.py` and LaTeX build:**

   (a) Calling `sp.series` on `H_g_full.subs(eta, lam_fluct * eta)` without `.doit()` caused a SymPy `RecursionError` on unevaluated `Derivative` nodes; fixed by calling `.doit()` before `sp.expand().coeff(lam_fluct, 2)`.

   (b) Integrating $\partial_U^2\omega_g(q;U)$ over $q\in[0,\infty)$ with symbolic $1-6U$ not declared positive returned a `Piecewise`; fixed by substituting a positive symbol $M_2 = 1-6U$.

(c) An initial transcription of the $O(U^3)$ and $O(U^4)$ Taylor coefficients of $W_T^B(U) - W_{b,\mathrm{ac}}(U)$ wrote $+\frac{9}{2\pi}U^3 - \frac{189}{16\pi}U^4$ instead of $+\frac{3}{\pi}U^3 - \frac{9}{2\pi}U^4$, and $\frac{1}{5}k^5$ instead of $\frac{2}{15}k^5$ for $\int u_{0,x}^2\,\mathrm{d}x$; caught and corrected by SymPy assertions.

(d) Direct `mp.quad` of the un-rationalised subtracted vacuum integrand on $[0,\infty)$ suffered catastrophic cancellation at large q, and the heat-kernel integral at $\delta = 10^{-8}$ required breakpoints at $\delta^{-1/2}$ and $4\delta^{-1/2}$; rationalising the integrand as $-18U^2q/[\sqrt{q^2+1}(\sqrt{q^2+1-6U}+\sqrt{q^2+1})^2]$ and adding the $\delta^{-1/2}$ breakpoints achieved $< 10^{-65}$ and $< 10^{-62}$ accuracy without relaxing any tolerance.

(e) The first `pdflatex` pass reported overfull boxes on un-broken filenames and the Section 4 table, plus `hyperref` bookmark math warnings; adding `\allowbreak`, `\footnotesize`, and `\texorpdfstring` resolved all warnings.